

Hydrodynamic Modelling & Control of Hydrofoil Vessels

Comparison of Optimal Control Methods: LQR & MPC

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School of Naval Architecture & Marine Engineering

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Outline

1 Motivation

- Hydrofoils
- Stability & Control
- Objectives

2 Hydrodynamic Modelling

- Rigid Body Dynamics
- Force Modelling

3 Linear System Analysis

- Linearization
- Linear System Characteristics

4 Optimal Control & Estimation

- Control Design
- State Estimation

5 Simulation Results

- Results

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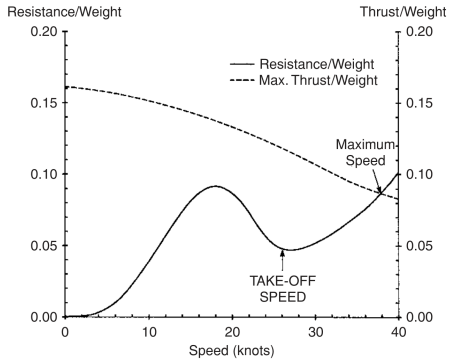
4 Optimal Control & Estimation

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Re-surfacing of Hydrofoils



Driven by

- IMO & EU regulations on fuel alternatives and zero emissions
- Minimal resistance at higher speed - fleet electrification



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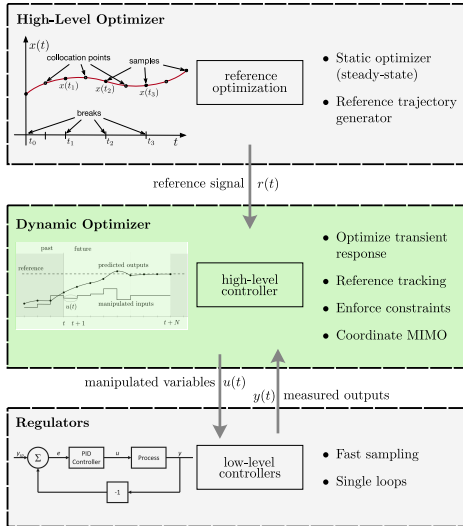
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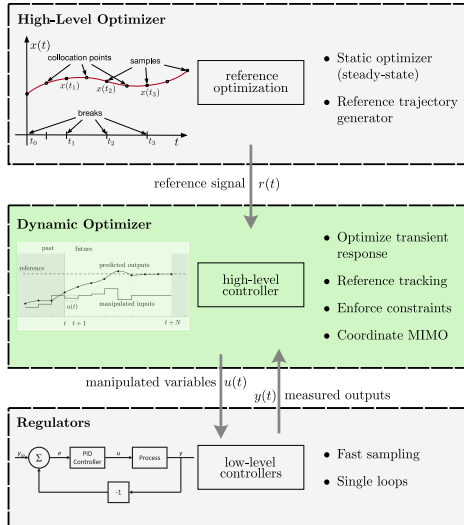
Control System Design



General control system architecture

- Trajectory generation produces reference $r(t)$
- *High level optimizer decides actuator setpoints $u(t)$*
- Simple SISO regulators track setpoints

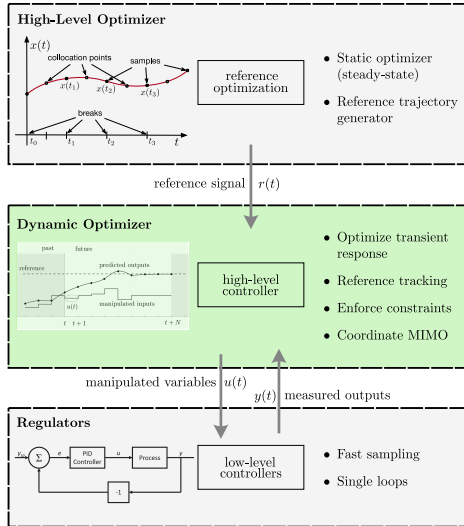
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What does this work attempt to solve?

- General, abstract modelling of a large class of vessels (hydrofoil type) based on parameters
- Automatic linearization at sequential operating points to assess system behavior
- Application of optimal control to mitigate stability issues apparent in fully submerged designs

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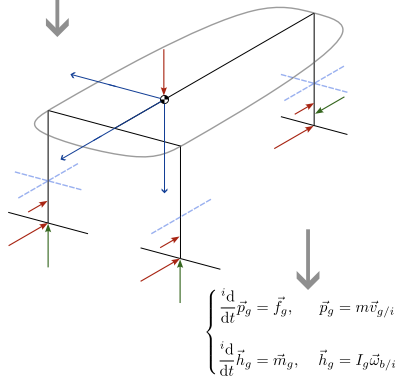
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Model Description



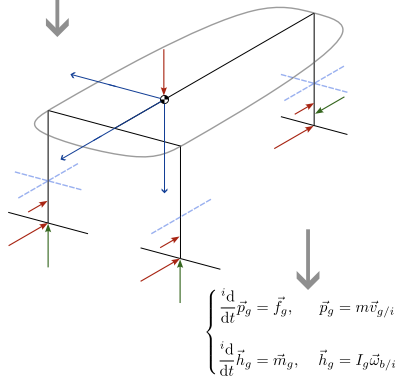
Constraints

- Configurable
- Computationally efficient
- Automatically linearizeable

Aiming to include

- Kinematics: Coordinate frames, motion geometry
- Kinetics: Main vessel as a rigid body
- Sea state generation
- Control surfaces
- Thrust producing devices

Model Description



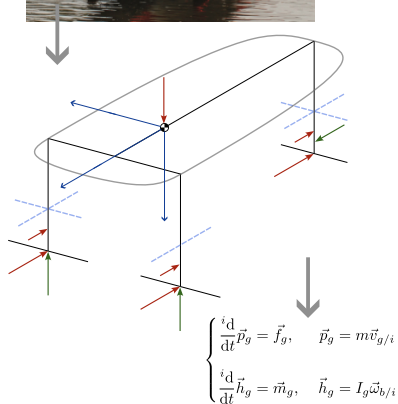
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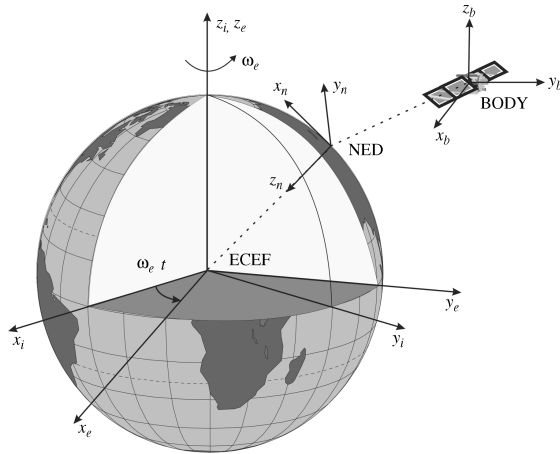
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Kinematics

Coordinate Frames & Reference Frames



- ECI $\{i\} = (x_i, y_i, z_i)$
- ECEF $\{e\} = (x_e, y_e, z_e)$
- NED $\{n\} = (x_n, y_n, z_n)$
- BODY $\{b\} = (x_b, y_b, z_b)$

CO as origin of $\{b\}$ as CG (may be time-dependent)

Kinematics

Motion Variables

$\{n\}$ position	$\mathbf{p}_{b/n}^n = \begin{bmatrix} N \\ E \\ D \end{bmatrix}$	Attitude	$\Theta_{nb} = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}$
$\{b\}$ linear velocity	$\mathbf{v}_{b/n}^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$	$\{b\}$ angular velocity	$\boldsymbol{\omega}_{b/n}^b = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$
$\{b\}$ force	$\mathbf{f}_b^b = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$	$\{b\}$ moment	$\mathbf{m}_b^b = \begin{bmatrix} K \\ M \\ N \end{bmatrix}$

- $\boldsymbol{\eta} = [N, E, D, \phi, \theta, \psi]^T$ generalized position-orientation vector
- $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ generalized velocity vector
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Kinematics & Kinetics

Framework Equations

Kinematic differential equations

Fossen's vectorial kinematics

$$\begin{bmatrix} \dot{\mathbf{p}}_{b/n}^n \\ \dot{\boldsymbol{\Theta}}_{nb} \end{bmatrix} = \begin{bmatrix} \mathbf{R}_b^n(\boldsymbol{\Theta}_{nb}) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{T}_{\boldsymbol{\Theta}}(\boldsymbol{\Theta}_{nb}) \end{bmatrix} \begin{bmatrix} \mathbf{v}_{b/n}^b \\ \boldsymbol{\omega}_{b/n}^b \end{bmatrix}$$

$$\dot{\boldsymbol{\eta}} = \mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta}) \boldsymbol{\nu}$$

Newton - Euler equations w.r.t CG are transformed to CO

Fossen's vectorial kinetics

$$\mathbf{M}_{RB} \dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu}) \boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

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- Matrix representation of EoM
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Non-linear simulation is performed by

$$\dot{\boldsymbol{\nu}} = \mathbf{M}_{RB}^{-1} (\boldsymbol{\tau}_{RB} - \mathbf{C}_{RB}(\boldsymbol{\nu}) \boldsymbol{\nu})$$

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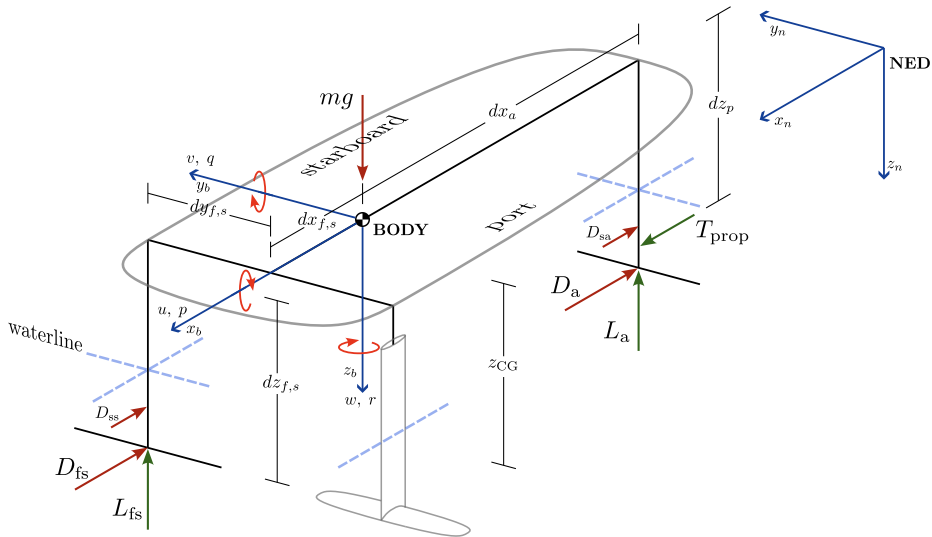
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The **force** vector

$$\boldsymbol{\tau} = \begin{bmatrix} \mathbf{f}^b \\ \mathbf{r}^b \times \mathbf{f}^b \end{bmatrix} = \begin{bmatrix} F_x \\ F_y \\ F_z \\ I_y F_z - I_z F_y \\ I_z F_x - I_x F_z \\ I_x F_y - I_y F_x \end{bmatrix}$$

Example: Gravitational force $\mathbf{f}_g^n = [0 \ 0 \ W]^T$
and moment arm $\mathbf{r}_g^b$



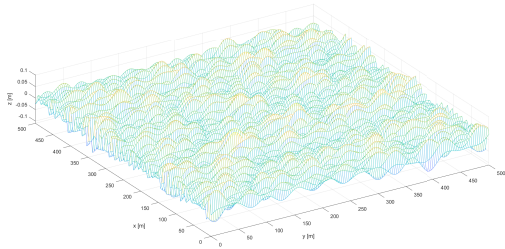
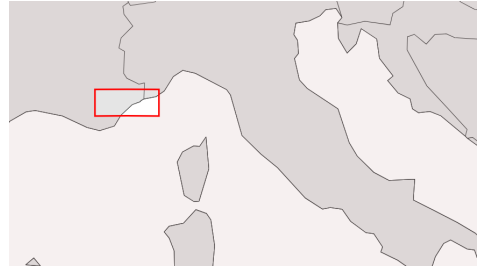
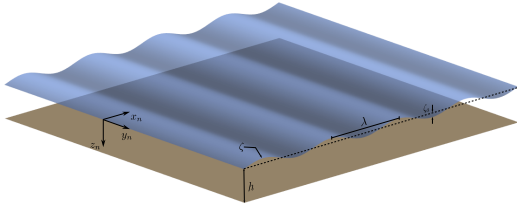
Frames
(Kinematics) and
modelled forces
(Kinetics)

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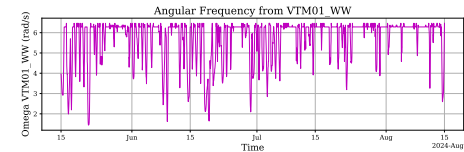
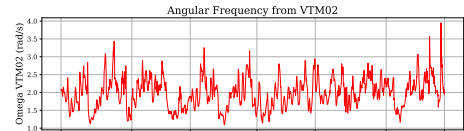
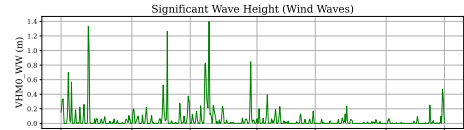
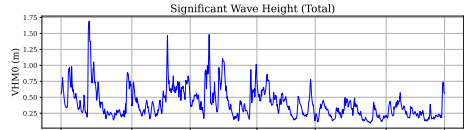
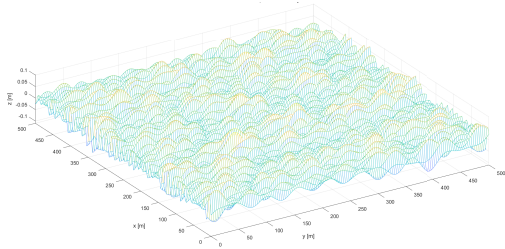
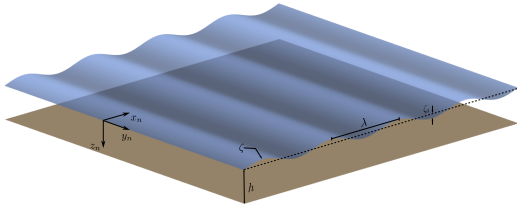
Environmental Forces

Waves



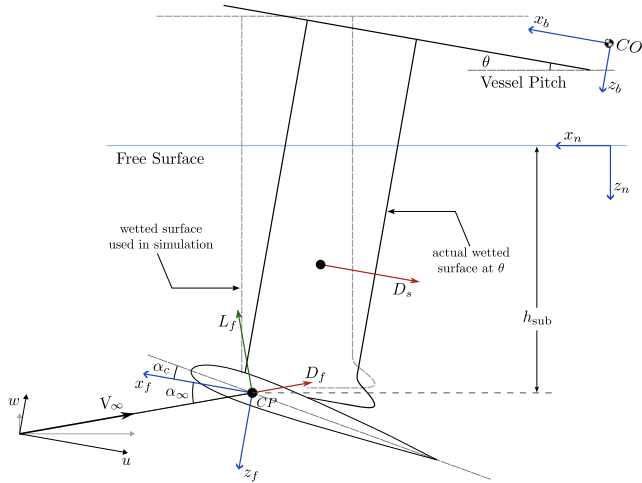
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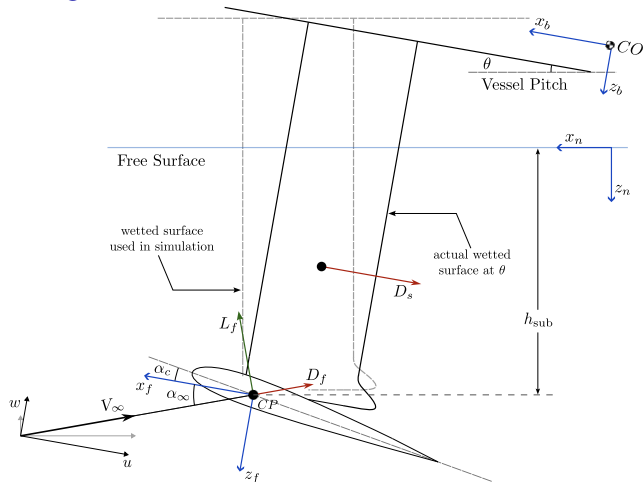
Actuation Forces

Lifting Surfaces



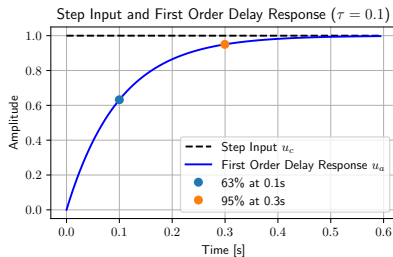
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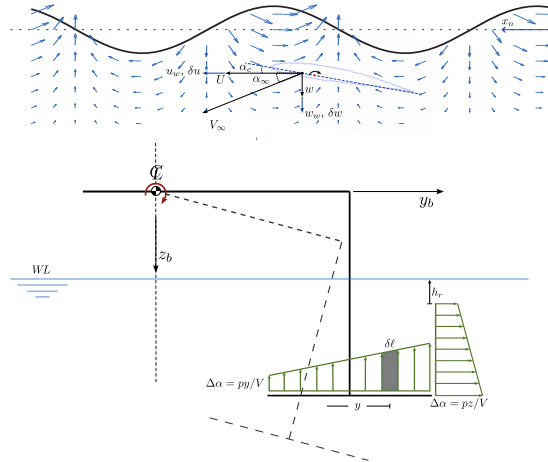
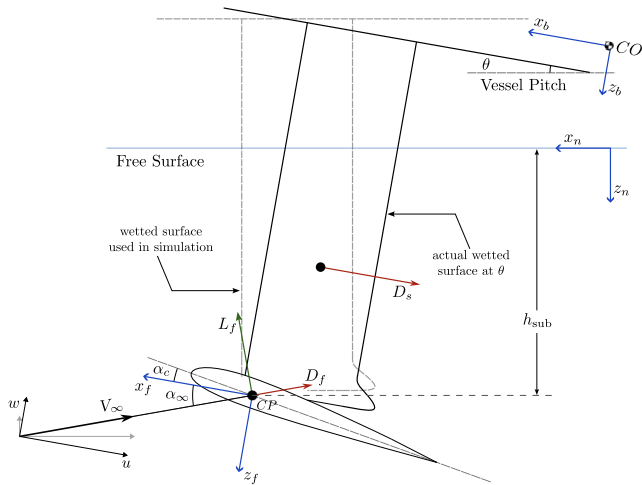
Actuators cannot respond *instantaneously*

$$g_f(s) = \frac{U_a(s)}{U_c(s)} = \frac{1}{\tau s + 1}$$



Actuation Forces

Lifting Surfaces

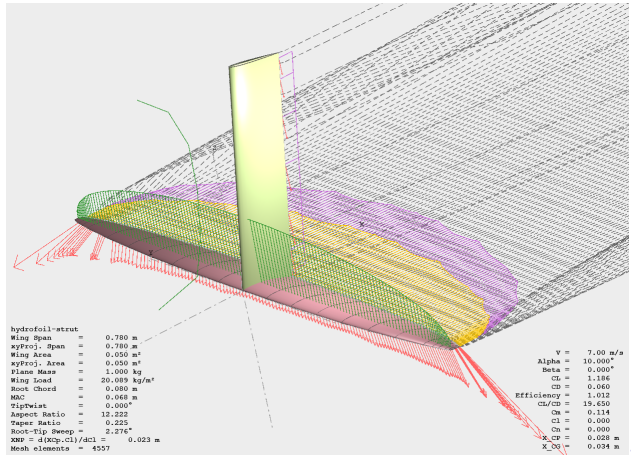
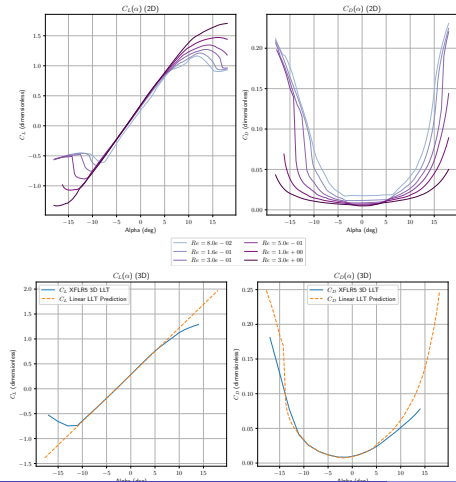


Forces – Hydrodynamic Analysis

Steady Lift

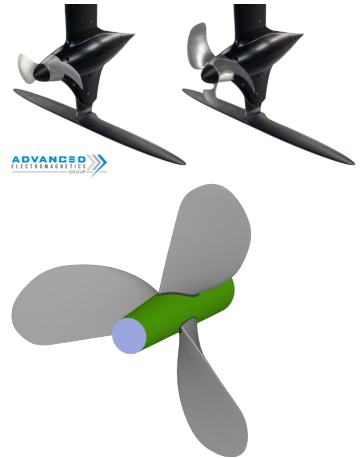
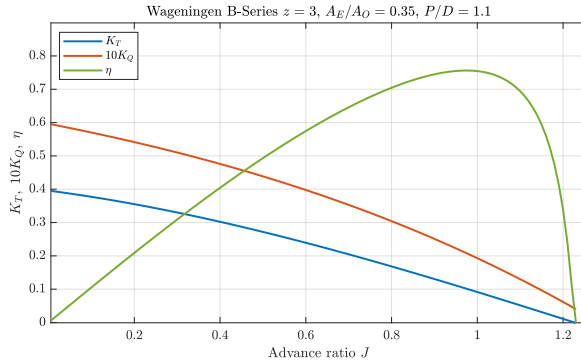
XFOIL data are interpolated by Re to account for 3D effects (XFLR5)

Overestimation in lift, *underestimation* in drag



Actuation Forces

Propulsion

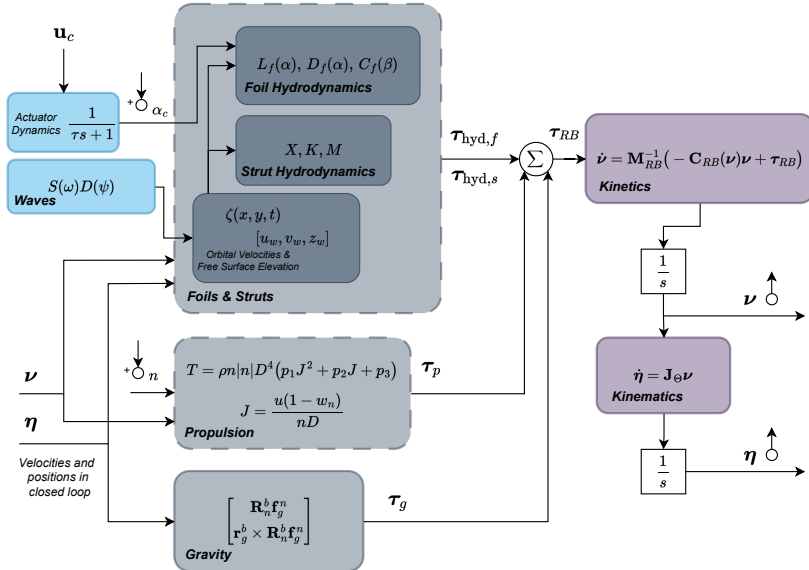


$$J = \frac{V_a}{nD}, \quad K_T = \frac{T_p}{\rho n D^4}, \quad K_Q = \frac{Q_p}{\rho n D^5}$$

$$T = \rho n |n| D^4 (p_{kT,1} J^2 + p_{kT,2} J + p_{kT,3})$$

$$Q = \rho n |n| D^5 (p_{kQ,1} J^2 + p_{kQ,2} J + p_{kQ,3})$$

Model Diagram



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Automatic Linearization

Find $\mathbf{u}_o = [\delta_w, \delta_w, \delta_t, T_p]$ for stable level flight $\mathbf{x}_o = [z_h, 0, 0, U_0, 0, 0, 0]$

Selection of states as $[z, \phi, \theta, w, p, q]$

- Find operating point (model trimming)
- Linearize around oper. point

$$\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}$$

$$\delta \mathbf{y} = \mathbf{C} \delta \mathbf{x} + \mathbf{D} \delta \mathbf{u}$$

Poles of linearized nominal model at $[-64.358, -15.489, -12.572, -1.615, 0, 0.196]$

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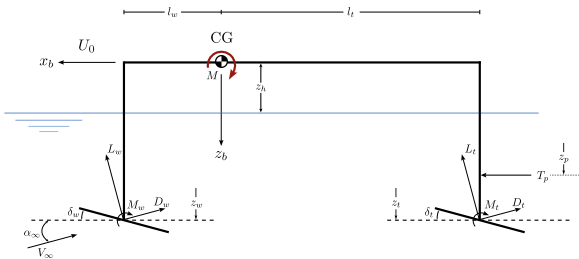
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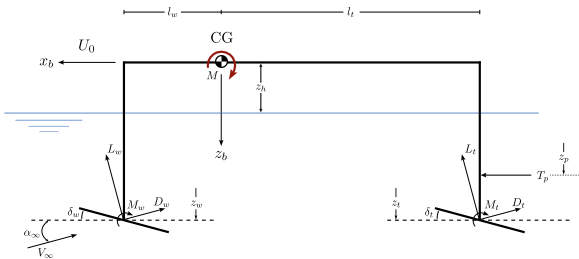
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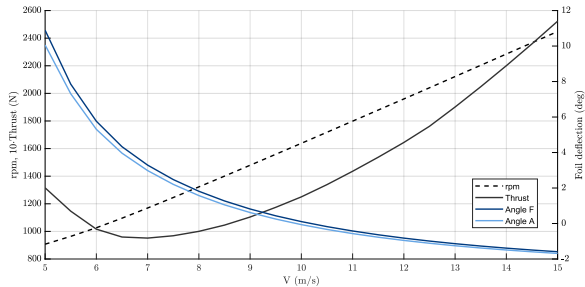
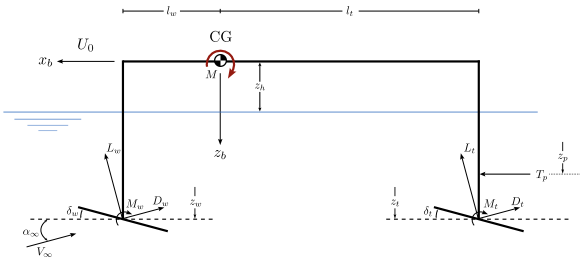
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- Linearize around oper. point

$$\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}$$

$$\delta \mathbf{y} = \mathbf{C} \delta \mathbf{x} + \mathbf{D} \delta \mathbf{u}$$



Automatic Linearization

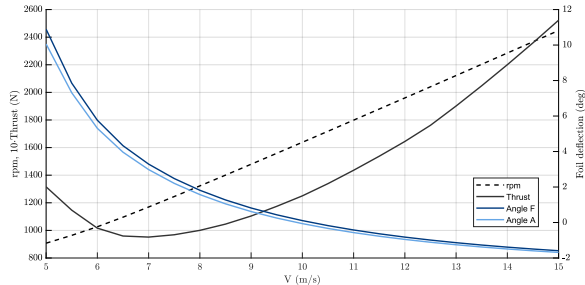
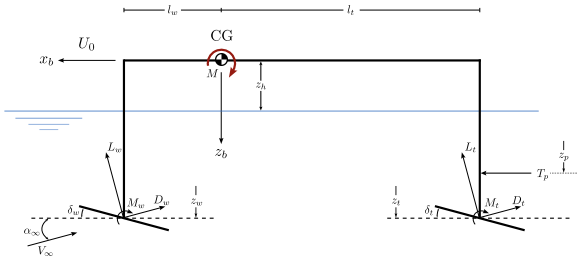
Find $\mathbf{u}_o = [\delta_w, \delta_w, \delta_t, T_p]$ for stable level flight $\mathbf{x}_o = [z_h, 0, 0, U_0, 0, 0, 0]$

Selection of states as $[z, \phi, \theta, w, p, q]$

- Find operating point (model trimming)
- Linearize around oper. point

$$\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}$$

$$\delta \mathbf{y} = \mathbf{C} \delta \mathbf{x} + \mathbf{D} \delta \mathbf{u}$$



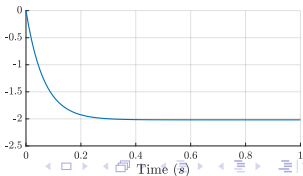
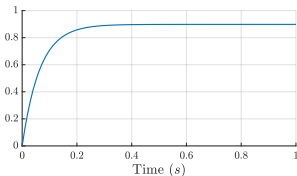
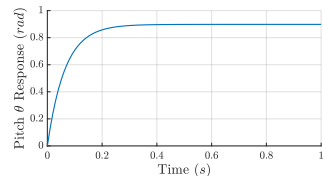
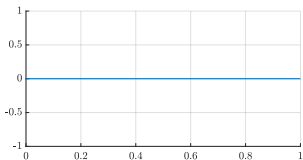
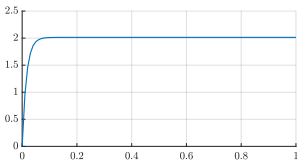
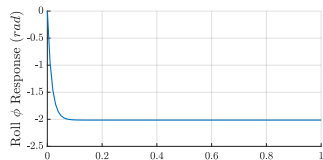
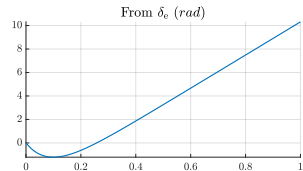
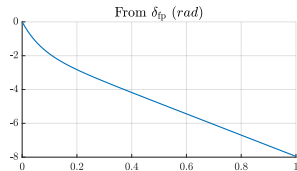
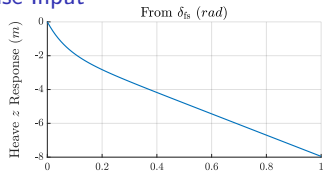
Poles of linearized nominal model at $[-64.358, -15.489, -12.572, -1.615, 0, 0.196]$

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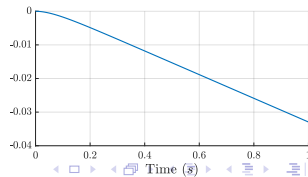
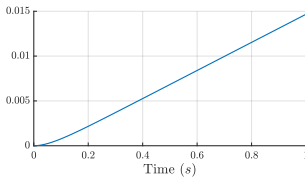
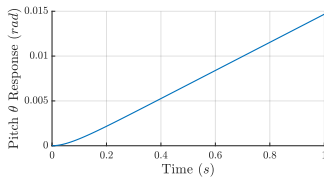
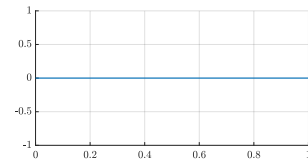
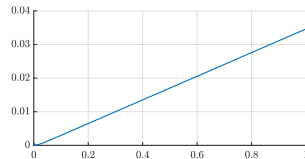
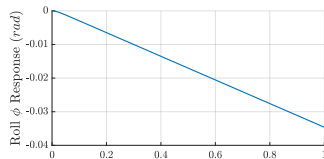
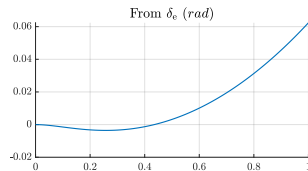
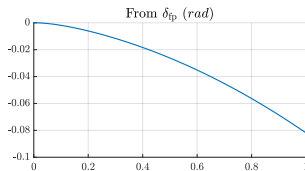
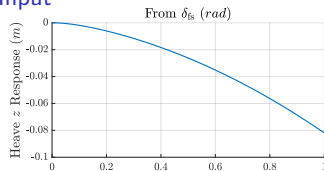
Actuation Response

Impulse Input



Actuation Response

Step Input

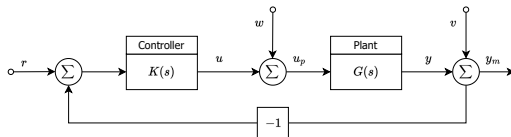


Outline

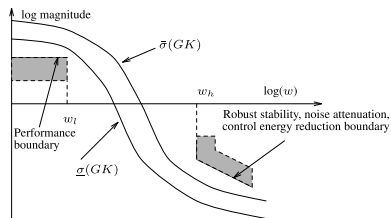
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Control Objectives & Performance Requirements

Controller design: Map $\mathbf{y} \mapsto \mathbf{u}$ given $\mathbf{r}$



Design Trade-offs (FD)



Objectives

- Stability
- Regulation and/or tracking
- Robustness
- Constraint Handling

$$\mathbf{y} = (\mathbf{I} + \mathbf{GK})^{-1} \underbrace{\mathbf{GK}}^L \mathbf{r} + \underbrace{(\mathbf{I} + \mathbf{GK})^{-1} \mathbf{G}}_S \mathbf{w} - \underbrace{(\mathbf{I} + \mathbf{GK})^{-1} \mathbf{GK}}_T \mathbf{v}$$

Discrete LQR & Integral Action

Augment standard LQR with integral states $\mathbf{x}_I = \int \mathbf{x} - \mathbf{r} \, dt$

Algorithm (dLQI)

Input: Reference r , State x

Output: Actuator command u

Parameters : $T_s, u_o, u^l, u^u, K, K_a$

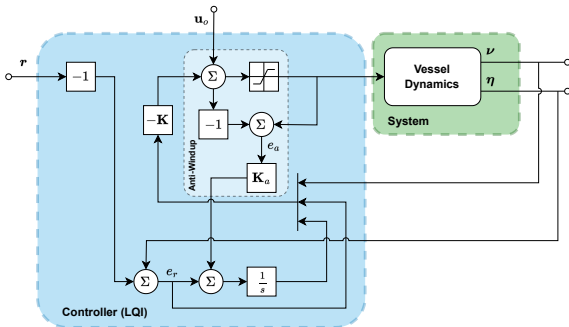
Persistent : x_I, u, u_c

```
1  if first call then
```

$$2 \quad \mathbf{x}_I, \mathbf{u}, \mathbf{u}_c \leftarrow 0$$
3 $\mathbf{e}_r \leftarrow \mathbf{C}_l \mathbf{x} - \mathbf{r}$
$$4 \quad \mathbf{x}_I \leftarrow \mathbf{x}_I + T_s (\mathbf{e}_r + \mathbf{K}_a (\mathbf{u} - \mathbf{u}_c))$$
$$5 \quad \mathbf{X}_{\text{aug}} \leftarrow [\mathbf{X}, \mathbf{X}_I]^\top$$
$$6 \quad \mathbf{u}_c \leftarrow \mathbf{u}_o - K \mathbf{x}_{\text{aug}}$$
$$7 \quad \mathbf{u} \leftarrow \min(\max(\mathbf{u}_c, \mathbf{u}^l), \mathbf{u}^u)$$

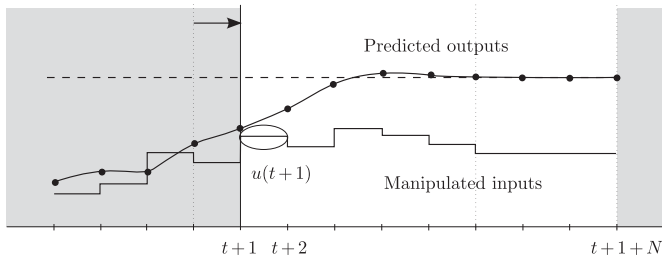
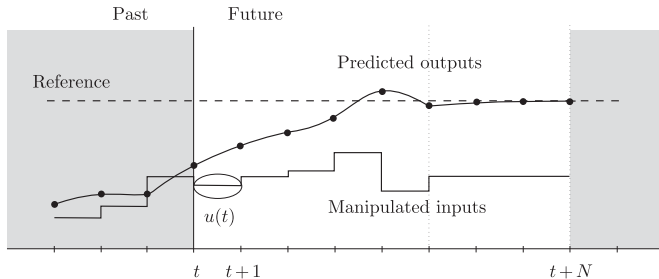
```
8 return  $u$ 
```

Block diagram



- LQI -S: Saturation
- LQI -W: Anti-windup

Receding Horizon Control Principle



MPC Algorithm at time t

- 1 Solve COCP over horizon N
- 2 Apply first ctrl action $\mathbf{U}^*(0)$
- 3 Shift horizon forward and repeat at $t+1$

MPC Structure

$$\mathbf{x}_c^\top = [\mathbf{x}_p^\top \mathbf{x}_{id}^\top \mathbf{x}_{od}^\top \mathbf{x}_n^\top]$$

$$\mathbf{u}_o^\top = [\mathbf{u}^\top \mathbf{v}^\top \mathbf{w}_{id}^\top \mathbf{w}_{od}^\top \mathbf{w}_n^\top]$$

MPC Implementation & State Estimation

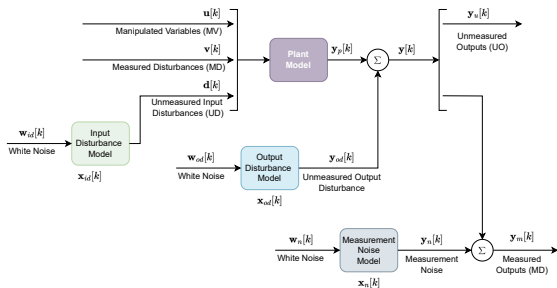
Cost Function Components: $J = J_y + J_u + J_{\Delta u} + J_\epsilon$

Output tracking, input penalties, move suppression, constraint softening

Extensions to standard linear MPC:

Input Disturbance [-I], Delay Prediction [-D], Reference Preview [-P], LPV Prediction Model

MPC Implementation & State Estimation



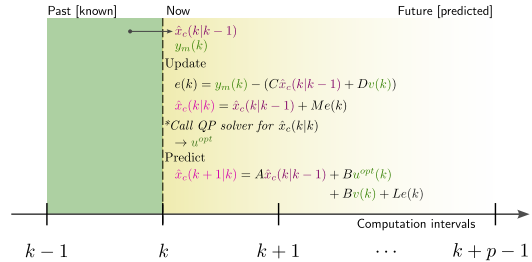
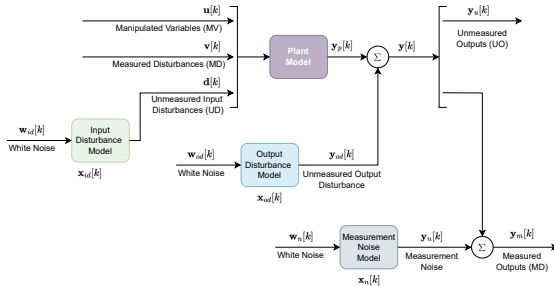
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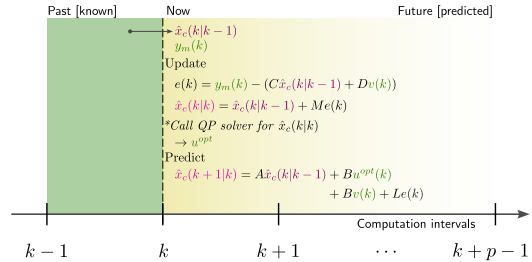
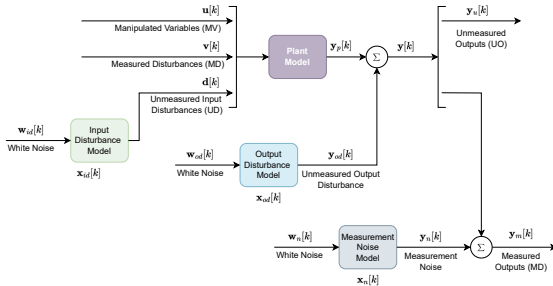
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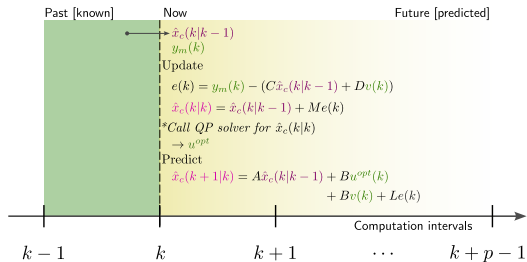
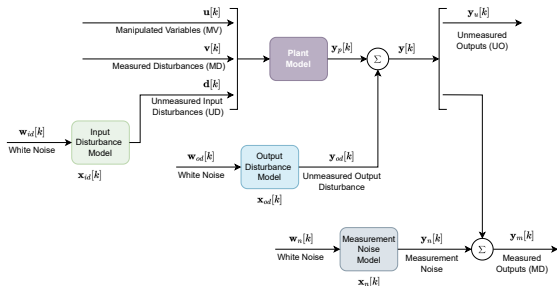
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Sensor Selection

No full state knowledge

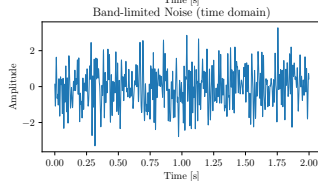
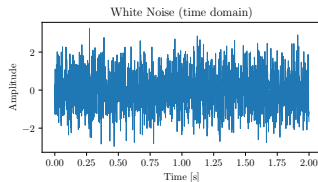
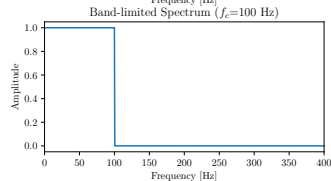
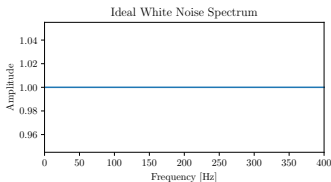
Fuse data sources (sensors) $\rightarrow$
reconstruct state vector (estimation)

- A GPS combined with an IMU (AHRS)
- An ultrasonic distance sensor

Available states: $\mathbf{v}^n, \boldsymbol{\omega}^b, \boldsymbol{\Theta}_{nb}$

And noisy height h_s

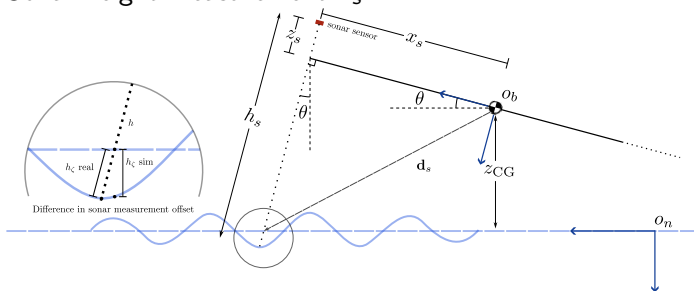
Modelling noise:



$$\sigma_n^2 = \int |e_n(f)H(f)|^2 df$$

Sensor Definitions – Estimator

Sonar height measurement h_s



Relationship between h_s and z_{CO}

$$z_{CO} = (R_b^n d_s^b)_3$$

Estimated states: $\mathbf{x} = [z, u, w]$

DKF Algorithm

Given inputs $\dot{\mathbf{u}}, \dot{\mathbf{w}}$

Prediction (a priori)

$$\hookrightarrow \hat{\mathbf{x}}_k^-, \mathbf{P}_k^-$$

Given measurements z_{CO}, V_{gps}

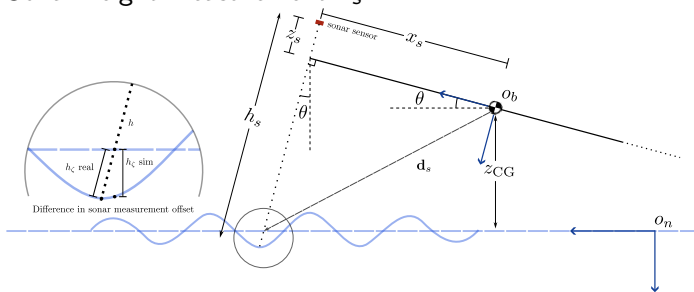
Correction (a posteriori)

$$\hookrightarrow \mathbf{K}_k, \hat{\mathbf{x}}_k^+, \mathbf{P}_k^+$$

Quantity	Units	$e_n [\cdot/\sqrt{\text{Hz}}]$	B(-3dB) [Hz]	σ (RMS)
Accel.	$\text{m} \cdot \text{s}^{-2}$	$5.9\text{e-}6$	500	$1.3\text{e-}2$
Angle	rad	—	—	$3.5\text{e-}3$
Ang. vel.	$\text{rad} \cdot \text{s}^{-1}$	$1.2\text{e-}4$	520	$2.8\text{e-}3$
Velocity	$\text{m} \cdot \text{s}^{-1}$	—	—	$5.0\text{e-}2$
Distance	m	—	≈ 20	$2.0\text{e-}3$

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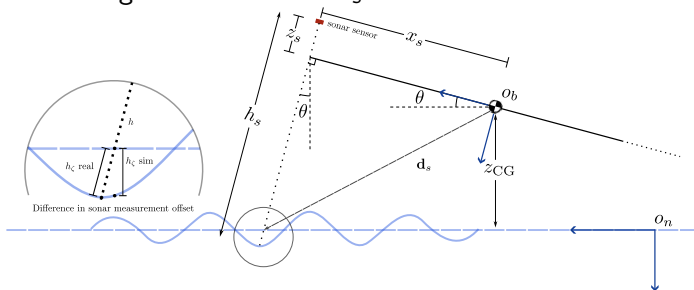
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Sonar height measurement h_s



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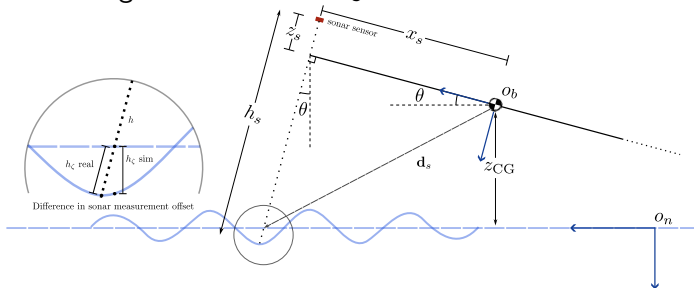
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Given measurements z_{CO}, V_{gps}

Correction (a posteriori)

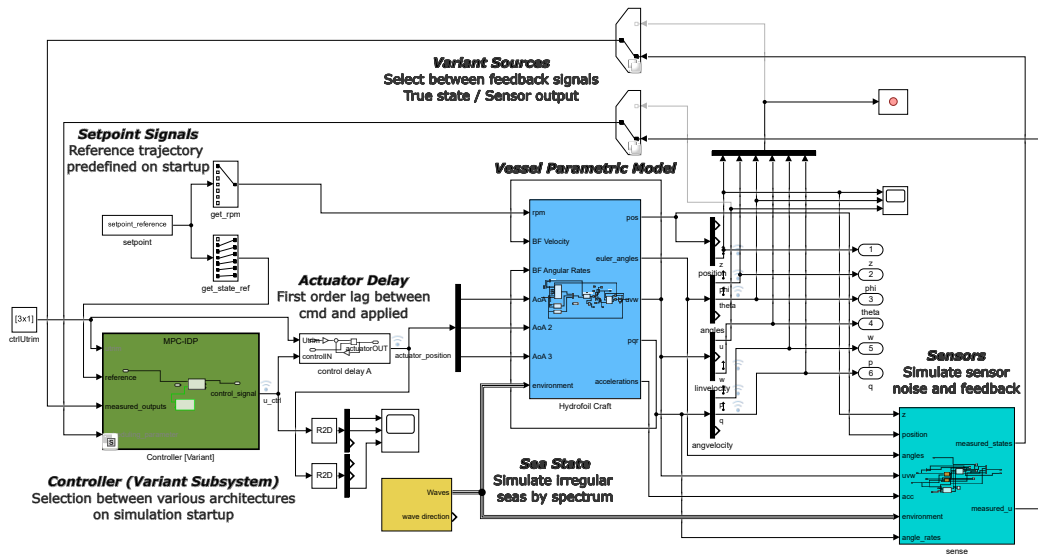
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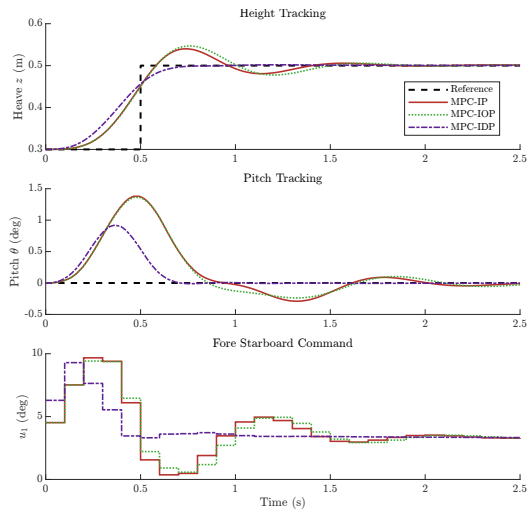
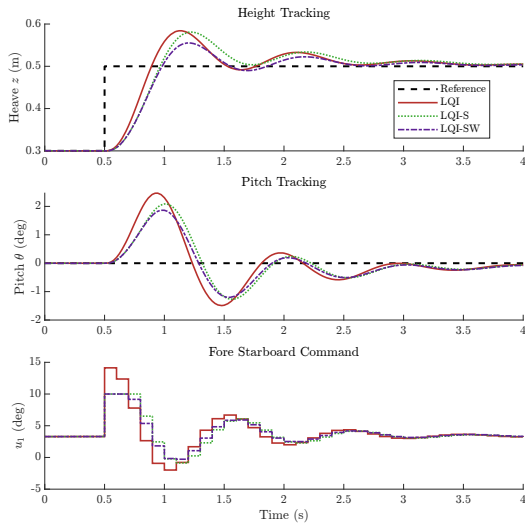
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Developed Simulation Environment



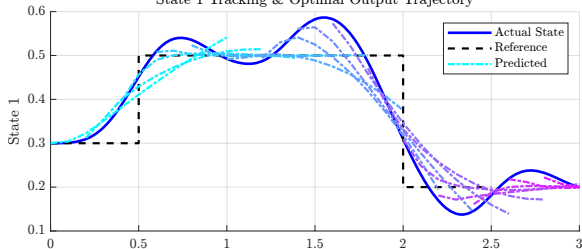
Reference Tracking



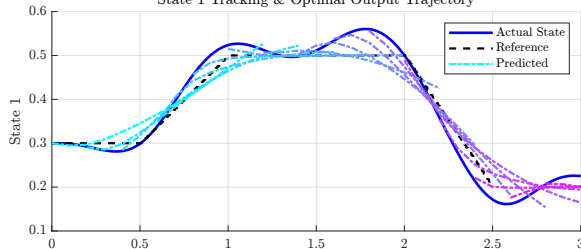
Performance metrics: ISE, IAE, ITAE & Actuator Effort

Up Close: Delay Prediction

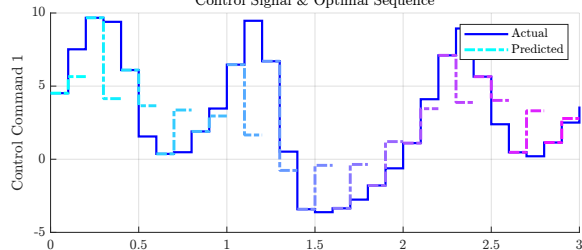
State 1 Tracking & Optimal Output Trajectory



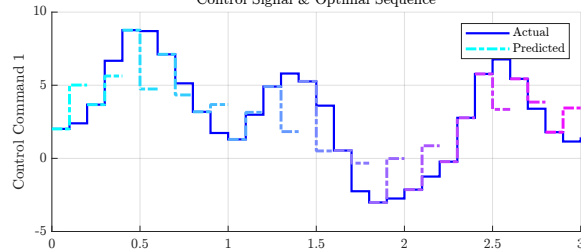
State 1 Tracking & Optimal Output Trajectory



Control Signal & Optimal Sequence

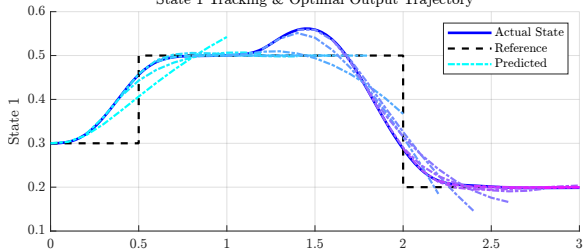


Control Signal & Optimal Sequence

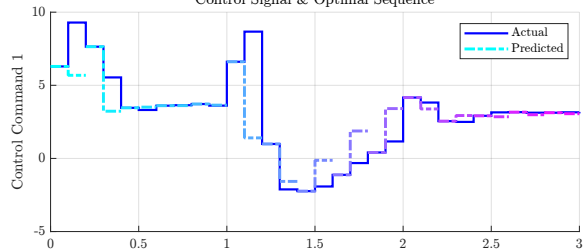


Up Close: Delay Prediction

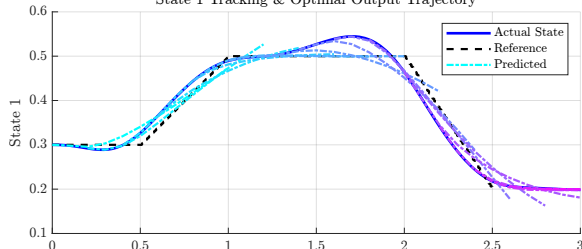
State 1 Tracking & Optimal Output Trajectory



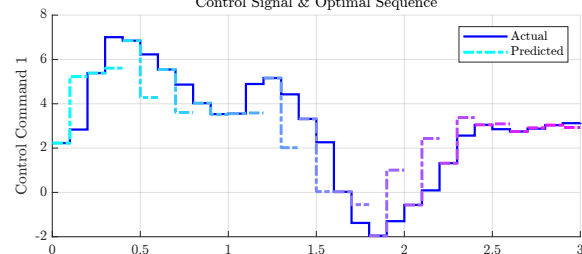
Control Signal & Optimal Sequence



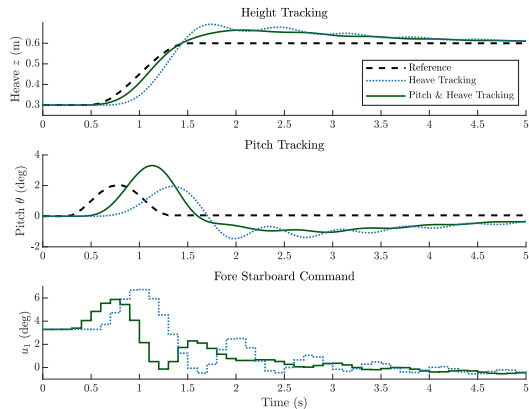
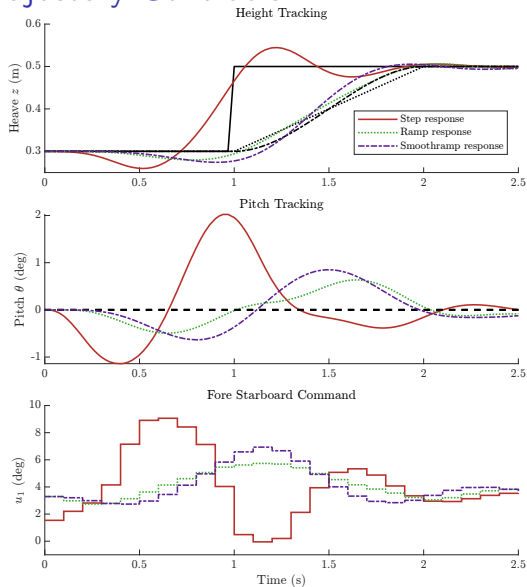
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Control Signal & Optimal Sequence

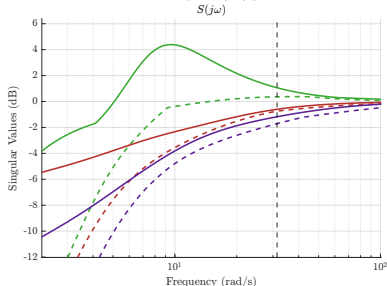
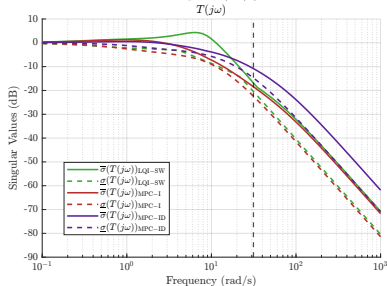
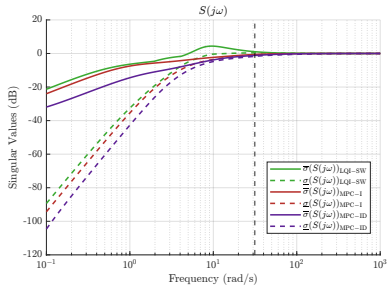
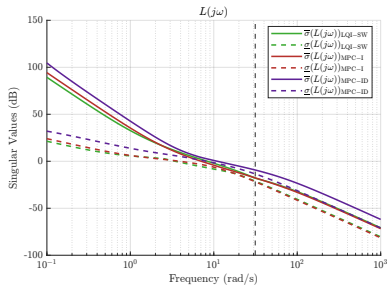


Trajectory Generation



Reference setpoints have an effect on performance

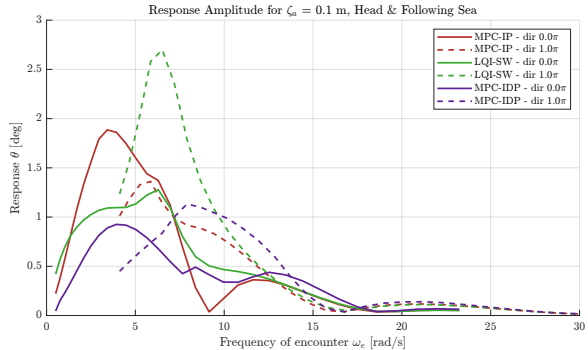
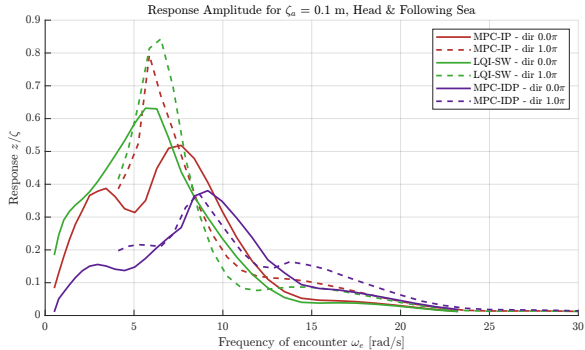
Disturbance Rejection



- L and ω_c in the desired regions
- S shows better tracking for MPC-IDP
- T classifies noise before Nyquist frequency

Wave Response

Encounter Frequency

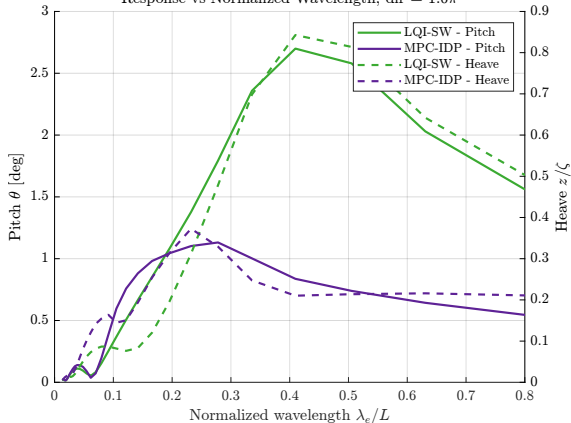


Visible regions of resonance – results on par with σ plots

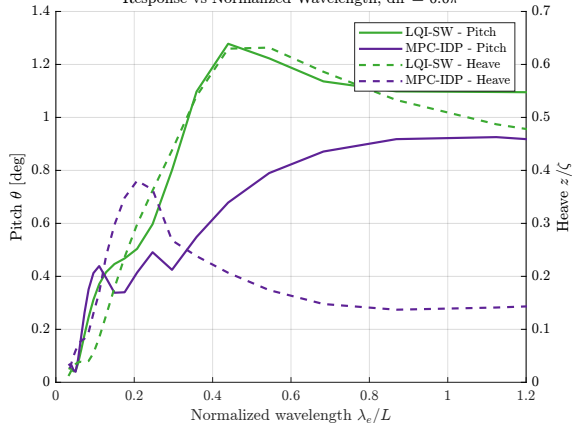
Wave Response

Encounter Lambda

Response vs Normalized Wavelength, dir = 1.0π

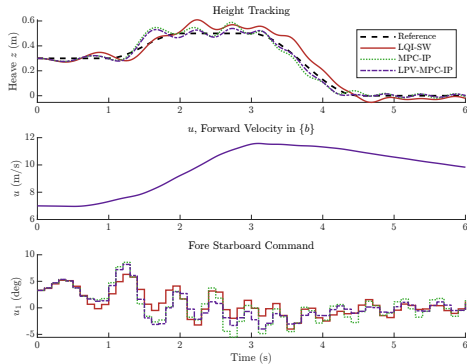


Response vs Normalized Wavelength, dir = 0.0π



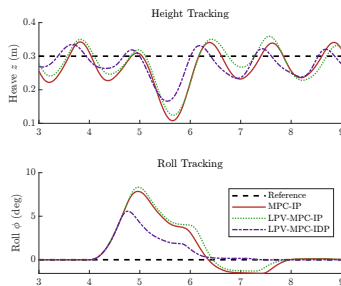
Additional Scenarios

Take-off & Touchdown

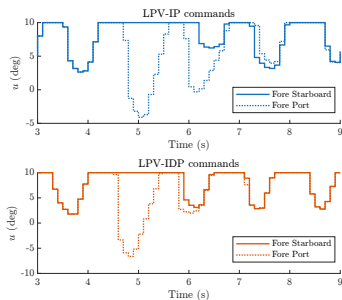


Reference tracking at the presence of wave and speed disturbance

Roll Corrections

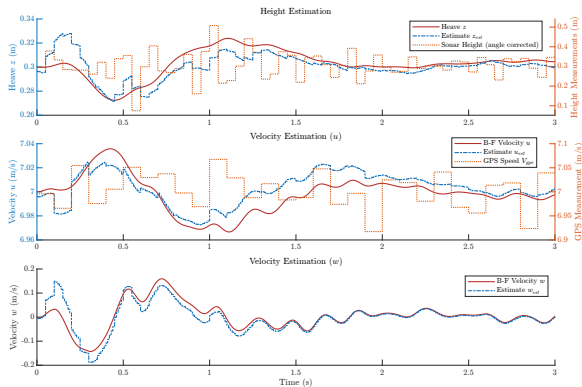


Ability to apply differential forces to reject roll disturbance

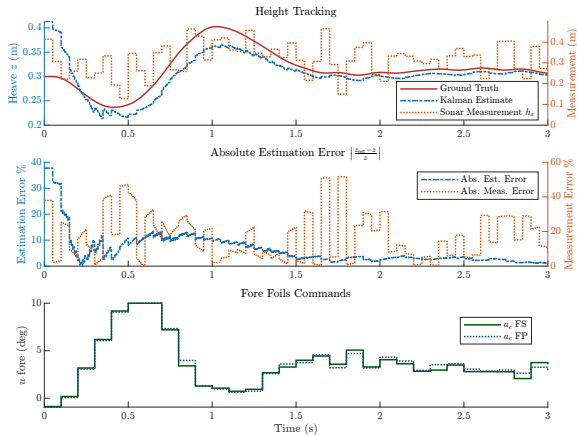


Estimation

All states are adequately reconstructed



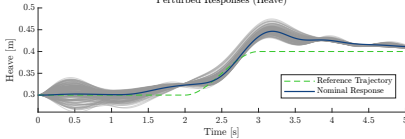
Estimator convergence depends on initialization



Model Uncertainty

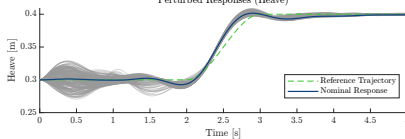
Parameter Sweep Response – LQI-SW

Perturbed Responses (Heave)

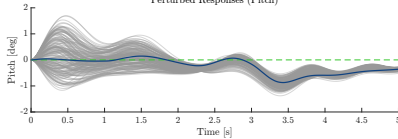


Parameter Sweep Response – LPV-MPC-IP

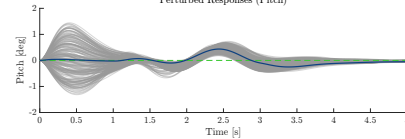
Perturbed Responses (Heave)



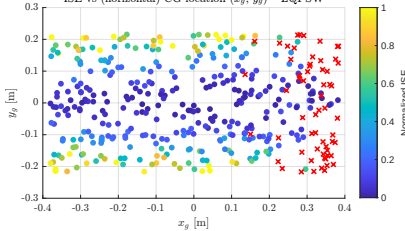
Perturbed Responses (Pitch)



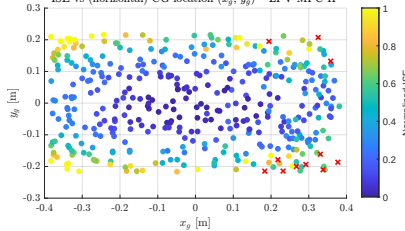
Perturbed Responses (Pitch)



ISE vs (horizontal) CG location (x_g, y_g) – LQI-SW



ISE vs (horizontal) CG location (x_g, y_g) – LPV-MPC-IP



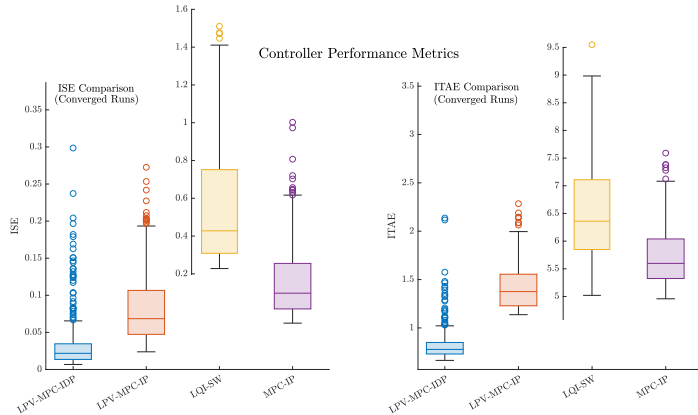
Response to perturbed models

- Keep designed controller
- Simulate $N = 400$ trials
- Mark unconverged runs
- Collect performance metrics

Perturbations in $m, \mathbf{r}_g^b, \mathbf{l}_g$

Statistical Assessment

- Failure rate for MPC 1.5–3%
- Failure rate for LQI 15%
- Significantly increased ISE & ITAE
- Delay prediction coupled with uncertainty – oscillations & outliers



Summary

- A parametric hydrodynamic model for a range of vessels is developed
- Automatic linearization and tools for system analysis
- Possible control configurations explored and compared against a baseline
- Outlook
 - ▶ The hydrodynamic model lacks accuracy in lateral DOF
 - ▶ No considerations of maneuvering

Future Work

Some possible extensions include

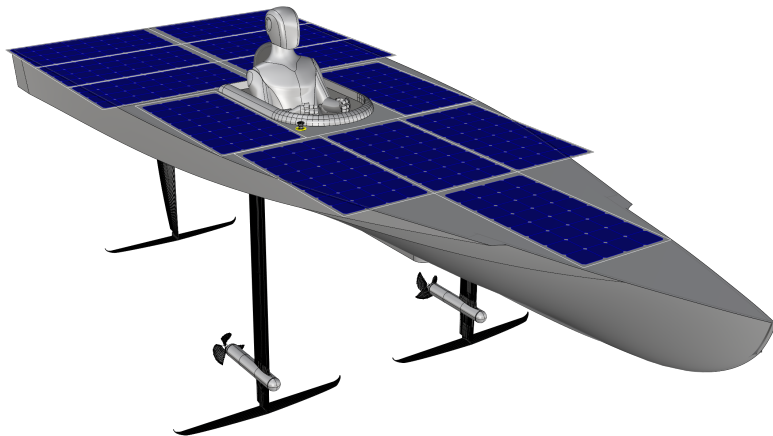
- Implementation of a non-linear dynamic lifting line model instead of Simulink + XFLR5
- Usage of a physics simulation engine that supports such applications (e.g Gazebo + ROS)
- Considerations for trajectory generation
- MPC extensions such as coupling with MHE, Tube-MPC etc

Given a platform for experimenting

- System identification, comparison to linearization output
- Implementation of algorithms on an embedded microprocessor
- Real time data collection for comparison to simulations

Future Work

Good news: A possible platform for experimenting



Thank You!

Questions?

Ioannis Skounakis Kounavis

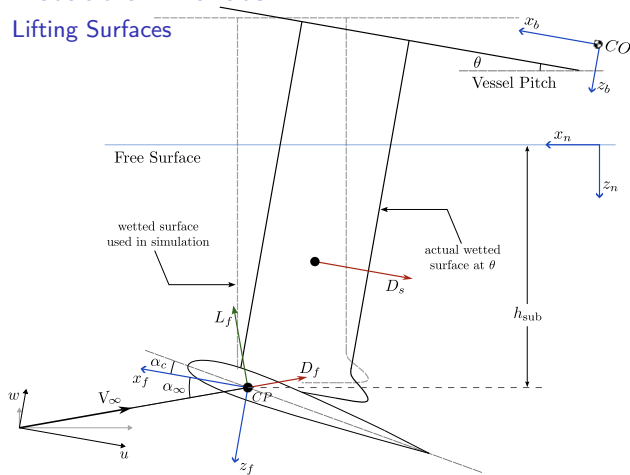
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Backup / Appendix

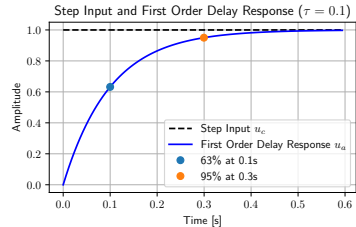
Actuation Forces

Lifting Surfaces



Actuators cannot respond *instantaneously*

$$g_f(s) = \frac{U_a(s)}{U_c(s)} = \frac{1}{\tau s + 1}$$



$$\mathbf{p}_{f/n}^n = \mathbf{p}_{b/n}^n + \mathbf{R}_b^n \mathbf{r}_f^b$$

$$h_{\text{sub}} = (\mathbf{p}_{f/n}^n)_3 + \zeta(\mathbf{p}_{f/n}^n)$$

$$\mathbf{f}_f^b = \begin{bmatrix} X_f \\ Y_f \\ Z_f \end{bmatrix} = \begin{bmatrix} \cos \alpha_\infty & 0 & -\sin \alpha_\infty \\ 0 & 1 & 0 \\ \sin \alpha_\infty & 0 & \cos \alpha_\infty \end{bmatrix} \begin{bmatrix} \cos \beta_\infty & -\sin \beta_\infty & 0 \\ \sin \beta_\infty & \cos \beta_\infty & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -D_f \\ -C_f \\ -L_f \end{bmatrix}$$

Unsteady Lift – Lateral Damping

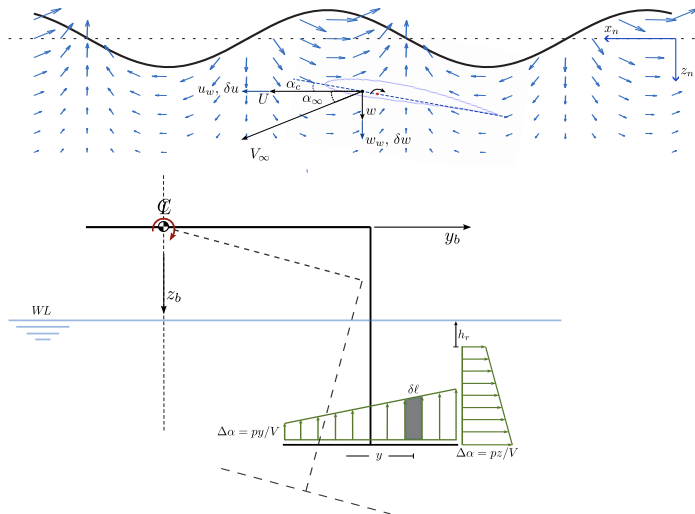
$$\mathbf{V}_{\infty}^f = \mathbf{v}_{b/n}^b + \boldsymbol{\omega}_{b/n}^b \times \mathbf{r}_f^b - \mathbf{R}_n^b \mathbf{v}_w^n$$

$$\alpha_{\infty} = \tan^{-1} \left(\frac{w_f}{u_f} \right)$$

$$\beta_{\infty} = \sin^{-1} \left(\frac{v_f}{\sqrt{u_f^2 + v_f^2 + w_f^2}} \right)$$

Eff. AoA: $\alpha_{\text{eff}} = \alpha_c + \alpha_\infty$

$$C_L^{2d} = C_{L,0} + C_{L_a} [\mathbb{C}(k_r)(\alpha_c + \alpha_b^{qs}) + \mathbb{S}(k_r)(\alpha_w^{qs})]$$



Model Reduction

State Selection & Linearization

Control requires lower DOF model

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))$$

Selection of states as $[z, \phi, \theta, w, p, q]$

LTI State Space

System defined as
perturbations from $(\mathbf{x}_o, \mathbf{u}_o)$

$$\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}$$

$$\delta \dot{\mathbf{y}} = \mathbf{C} \delta \mathbf{x} + \mathbf{D} \delta \mathbf{u}$$

EOM are reduced to (nominal)

Body-fixed rates

$$\dot{u} = -\frac{qw}{m} + \frac{X_{\text{hyd}}}{m} - g \sin \theta$$

$$\dot{w} = \frac{qu}{m} + \frac{Z_{\text{hyd}}}{m} + g \cos \theta \cos \phi$$

$$\dot{p} = \frac{K}{I_x}$$

$$\dot{q} = \frac{M}{I_y}$$

NED rates

$$\dot{z} = -u \sin \theta + w \cos \phi \cos \theta$$

$$\dot{\phi} = p + q \sin \phi \tan \theta$$

$$\dot{\theta} = q \cos \phi$$

Extensions to the Standard MPC Scheme

Input Disturbance [-I]

$$\begin{aligned}\mathbf{x}_{k+1} &= \mathbf{A}_p \mathbf{x}_k + \mathbf{B}_{pu} \mathbf{u}_k + \mathbf{B}_{pd} \mathbf{d}_{id,k} \\ \mathbf{d}_{id,k} &= \mathbf{C}_{id} (\mathbf{A}_{id} \mathbf{x}_{id,k-1} + \mathbf{B}_{id} \mathbf{w}_{id,k-1})\end{aligned}$$

Adapt to constant disturbances in inputs (lifting surface coefficients)

Delay Prediction [-D]

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{u}}_a \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ 0_{m \times n} & -\frac{1}{\tau} \mathbf{I}_m \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u}_a \end{bmatrix} + \begin{bmatrix} 0_{n \times m} \\ \frac{1}{\tau} \mathbf{I}_m \end{bmatrix} \mathbf{u}_c$$

Account for delayed actuator response

Reference Preview [-P]

$$\min_{\mathbf{z}} \sum_{i=0}^{p-1} \left\| \mathbf{L}_{Sy}^{-1} \mathbf{W}^y (\mathbf{y}_{i+1} - \mathbf{r}_{k+i}) \right\|_2^2$$

Anticipative action

LPV Prediction Model

$$\mathbf{x}[k+1] = \mathbf{A}_p(\boldsymbol{\lambda}(k)) \mathbf{x}_p[k] + \mathbf{B}_{pu}(\boldsymbol{\lambda}(k)) \mathbf{u}[k]$$

Adapt to changing dynamics by array of linearizations (constant over horizon)

Quantifying Performance

$$\text{ISE: } \int_{t_0}^{t_f} \sum_i |e_i(t)|^2 dt = \|\mathbf{e}(t)\|_2^2$$

$$\text{IAE: } \int_{t_0}^{t_f} \sum_i |e_i(t)| dt = \|\mathbf{e}(t)\|_1$$

$$\text{ITAE: } \int_{t_0}^{t_f} t \sum_i |e_i(t)| dt$$

$$\text{Actuator Effort: } \int_{t_0}^{t_f} t \sum_i |u_i(t)|^2 dt = \|\mathbf{u}(t)\|_2^2$$

Available tools for assessing performance

- Time Domain: Performance Metrics
- Frequency Domain: Singular Values
- Probabilistic Methods